Axiom at AraSeg Shared Task 2026: A Regularized Dual-Model Ensemble with Domain-Adaptive Pretraining for Arabic Sentence Segmentation

Md. Ajwad Hossain[†]
[†]Chittagong University of Engineering & Technology
md.ajwadhossain@gmail.com

Abstract

Identifying sentence boundaries in Arabic text is a challenging token classification task due to extreme dialectal variations, the absence of rigid punctuation rules, and document lengths that frequently exceed the capacity of standard transformer models. In this paper, we describe Team Axiom's submission to the Paragraph-Aware Open track of the AraSeg Shared Task 2026. We propose a highly regularized, dual-model ensemble combining MARBERTv2 and CAMeLBERT-MSA. To address the challenge of excessively long documents, we employ a sliding-window tokenization strategy. Furthermore, we adapt our base models to the task corpus using Domain-Adaptive Pretraining (DAPT). Our final system achieved a Macro-F1 score of 90.2% (Precision: 91.9%, Recall: 89.5%), securing 7th place in the shared task.

1 Introduction

Sentence Boundary Detection (SBD) is a foundational preprocessing step for downstream Natural Language Processing (NLP) tasks such as machine translation and document summarization. While SBD is often treated as a solved problem in resource-rich languages heavily reliant on terminal punctuation, Arabic presents a unique set of linguistic challenges.

Problem Statement: Arabic text frequently lacks rigid punctuation rules, relying heavily on coordinating conjunctions (e.g., *wa* / "and", *fa* / "then") to connect continuous, multipage clauses. Furthermore, the corpus provided for this shared task spans highly divergent genres and dialects, introducing severe semantic ambiguity at clause boundaries. Standard transformer architectures, which are structurally constrained by a quadratic attention mechanism to a 512-token limit (Beltagy et al., 2020), are fundamentally unsuited for this dataset. Standard truncation would result in a catastrophic loss of contextual boundaries.

To solve this, we propose a problem-specific pipeline. We circumvent the sequence bottleneck using an overlapping sliding-window tokenization strategy. To combat the severe dialectal variance, we apply Domain-Adaptive Pretraining (DAPT) (Gururangan et al., 2020) to adapt the model weights to the target corpus. Finally, recognizing that single models struggle with the high variance of Arabic conjunctions, we deploy a highly regularized, dual-model ensemble of MARBERTv2 (Abdul-Mageed et al., 2021) and CAMeLBERT-MSA (Inoue et al., 2021) to achieve mathematically stable boundary predictions.

2 Background and Related Work

2.1 Task Formulation

We formulate Arabic sentence segmentation as a supervised binary token classification task. Let a document be a sequence of tokens $T = [t_1, t_2, ..., t_n]$. The model learns a mapping $f(T) \rightarrow Y$, where $Y = [y_1, y_2, ..., y_n]$ and $y_i \in \{0, 1\}$. A label of 1 indicates that t_i is the final token of a sentence. Our system was submitted to the Paragraph-Aware Open Track of the AraSeg Shared Task 2026 (Elkholy et al., 2026b), utilizing the official AraSeg Corpus (Elkholy et al., 2026a).

2.2 Linguistic Challenges: An Input/Output Example

Unlike clean Modern Standard Arabic (MSA), dialectal and multi-genre Arabic frequently exhibits sparse or missing punctuation (Yagi et al., 2024). Consequently, segmenters that over-rely on punctuation lose robustness (Frohmann et al., 2024). The most prevalent ambiguity arises from coordinating conjunctions, notably the prefix *wa-* ("and"), which can act as both an intra-sentence linker and a hard sentence boundary.

Table 1 illustrates this challenge, demonstrating how the model must utilize deep contextual seman-

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tics rather than relying on explicit punctuation or simple conjunction rules to assign the boundary label ($y_i = 1$).

Token (t_i)	Transliteration	Label (y_i)
وصل	wasala (arrived)	0
القطار	al-qitar (the train)	0
متأخرا	muta'akhiran (late)	1
و	**wa- (and/so)**	**0**
لذلك	lithalika (therefore)	0
قررنا	qararna (we decided)	0

Table 1: An input/output example illustrating boundary ambiguity. The conjunction *wa-* acts as a new sentence starter, making the preceding token (muta'akhiran) the boundary.

2.3 Handling Long Sequences

The absolute position embeddings and self-attention complexity of BERT-based models restrict inputs to 512 subwords (Devlin et al., 2019). For document-level token classification, previous works have demonstrated that segmenting text into independent, non-overlapping chunks destroys cross-sentence context at the chunk boundaries. We follow the established literature by employing a sliding-window approach with dynamic strides, allowing overlapping boundary tokens to retain bidirectional context across chunk margins.

3 System Overview

Figure 1 illustrates the complete architecture of our proposed pipeline. To address the severe dialectal variance and document length constraints of the AraSeg corpus, our system is structured into three core stages: context-aware tokenization to preserve bidirectional context, domain-adaptive parallel encoding using dual transformer backbones, and a calibrated logit fusion mechanism to generate mathematically stable boundary predictions.

3.1 Domain-Adaptive Pretraining (DAPT)

Pretrained language models often struggle when applied to specialized domains that differ from their original pretraining corpora. Following the methodology of Gururangan et al. (2020), we applied DAPT to both MARBERTv2 and CAMeLBERT-MSA prior to fine-tuning on the classification task. We performed Masked Language Modeling (MLM) using the unlabelled training data, allowing the models to adapt to the specific dialectal structures and punctuation distributions of the AraSeg corpus.

3.2 Context-Aware Tokenization

Corpus analysis revealed that 63.5% of the training documents exceeded 512 subwords, with a maximum length reaching 22,692 subwords. To prevent the loss of contextual boundaries through truncation, we implemented a sliding-window tokenization strategy with a stride of 64 tokens.

During subword tokenization, alignment artifacts frequently occur. To ensure the loss function accurately penalized the model, we applied a strict subword label masking technique: the true label (0 or 1) was assigned only to the first subpiece of a word, while all subsequent subpieces and special tokens (e.g., `[CLS]`, `[SEP]`) were masked with a value of `-100`. PyTorch's Cross-Entropy loss inherently ignores indices with this value, forcing the model to optimize strictly on semantic word boundaries.

3.3 The Dual-Model Ensemble

Different pretraining data produce varying downstream strengths, and aggregating models with complementary biases effectively reduces variance (Alammary, 2024). To leverage this, our final architecture is a dual-model ensemble relying on DAPT-adapted MARBERTv2 and CAMeLBERT-MSA.

To resolve the overlapping token predictions generated by our sliding-window strategy, we first average the logit outputs for any token t_i across all chunks C that contain it (Jaiswal and Milios, 2023), defined as:

$$\hat{P}_{model}(y_i = 1) = \frac{1}{|C|} \sum_{c \in C} P_c(y_i = 1 | t_i) \quad (1)$$

Through extensive hyperparameter tuning on the development set, we found the optimal balance by slightly favoring the CAMeLBERT predictions. The final ensemble prediction is calculated via a weighted sum, and a calibrated threshold ($\tau = 0.5130$) is applied to output the binary prediction:

$$P_{final}(y_i = 1) = 0.5208 \cdot \hat{P}_{CAM}(y_i = 1) + 0.4792 \cdot \hat{P}_{MAR}(y_i = 1) \quad (2)$$

Lastly, we implement a paragraph-aware heuristic: the model automatically forces a boundary prediction ($y = 1$) for the absolute final token of

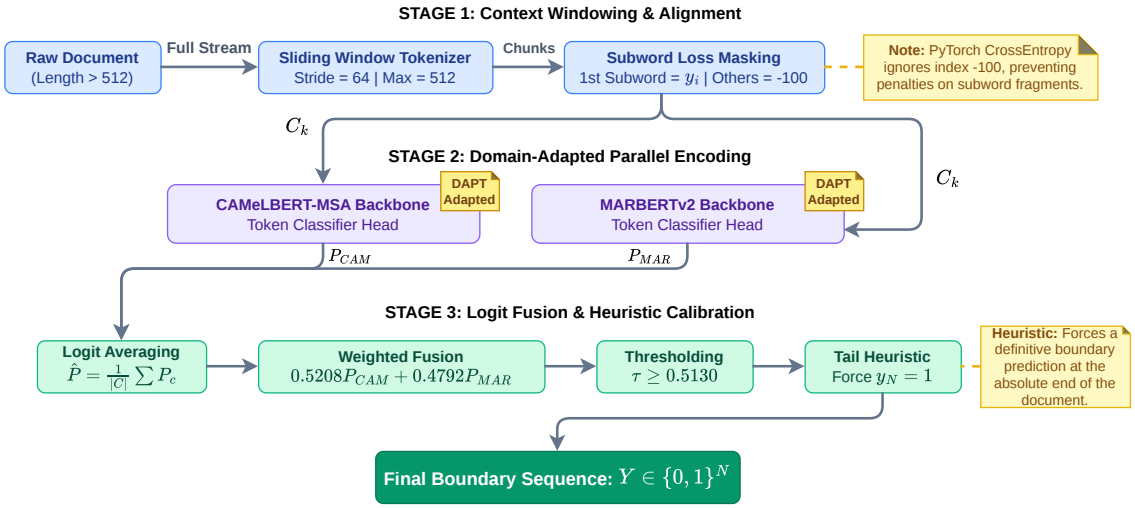


Figure 1: Overview of system architecture: (1) Sliding-window tokenization with subword label masking; (2) Parallel domain-adapted language model encoding; (3) Chunk-wise logit aggregation, weighted ensemble fusion, calibrated thresholding, and paragraph-aware boundary calibration.

any document, ensuring a boundary that neural inference occasionally misses due to a lack of trailing context.

4 Experimental Setup

4.1 Data

The system was trained and evaluated utilizing the official AraSeg Corpus. To rigorously tune our ensemble weights and decision thresholds, we reserved a subset of the official training data (18 documents) as our internal development set, leaving the remaining 156 documents for training. Final system evaluation was conducted on the official 100-document Blind Test set provided via CodaBench for the Paragraph-Aware Open Track. Detailed length statistics for these splits are provided in Appendix B.

4.2 Hyperparameters and Infrastructure

To manage the memory footprint of the transformer architectures on limited GPU hardware, gradient accumulation was utilized during both DAPT and fine-tuning. Detailed hyperparameters are provided in Appendix A.

5 Results and Analysis

5.1 Official Results

As shown in Table 2, our dual-model ensemble achieved an F1-score of 90.2%. The system exhibited a strong Precision (91.9%), indicating that the regularization of the dual-ensemble successfully mitigated the base models' tendency to aggressively over-predict boundaries around ambiguous coordinating conjunctions.

Team	Precision	Recall	F1
Axiom (Ours)	91.9	89.5	**90.2**

Table 2: Official results on the AraSeg Blind Test set.

5.2 Error Analysis: The Cost of Weighted Loss

While our baseline ensemble achieved strong results, our error analysis revealed a critical trade-off regarding recall optimization. Recognizing that the dataset heavily favors the non-boundary class ($y_i = 0$), we experimented with cost-sensitive learning, applying a heavily weighted cross-entropy loss function ($w_0 = 0.1, w_1 = 0.9$) during fine-tuning.

Our empirical observations showed that while this approach artificially forced the model to catch hard-to-find boundaries (pushing validation recall to 91.7%), it resulted in a severe degradation of precision. The model became hypersensitive to coordinating conjunctions, frequently predicting boundaries at every occurrence of *wa-* or *fa-*, regardless of context. Ultimately, standard unweighted fine-tuning combined with our dual-ensemble probability averaging provided the most mathematically stable predictions on unseen data.

6 Conclusion

In this paper, we detailed Team Axiom's approach to the Paragraph-Aware Open Track of the AraSeg 2026 Shared Task. We demonstrated that the architectural limitations of standard transformers on long documents can be effectively mitigated through an overlapping sliding-window tokenization strategy. Furthermore, our results indicate that coupling Domain-Adaptive Pretraining (DAPT) with a probability-averaged dual-encoder ensemble (MARBERTv2 and CAMeLBERT-MSA) stabilizes predictions against highly ambiguous Arabic coordinating conjunctions. This heavily regularized pipeline achieved a Macro-F1 score of 90.2%, proving highly robust on unseen multi-genre data. Future work will explore dynamic decision thresholding and deeper morphological analysis to further reduce false-positive boundary predictions without sacrificing recall.

Limitations

While our dual-model ensemble effectively mitigates boundary ambiguity, it introduces a high computational burden. Processing documents with an overlapping sliding window effectively multiplies the total number of tokens evaluated. When combined with the parameter footprint of running both MARBERTv2 and CAMeLBERT-MSA simultaneously, the inference latency makes the system sub-optimal for real-time segmentation tasks. Additionally, our error analysis indicated that while the system achieves high precision, attempts to boost recall via cost-sensitive learning resulted in a severe degradation of precision due to hypersensitivity to Arabic coordinating conjunctions. Future iterations must address this recall bottleneck without relying heavily on computationally expensive ensemble averaging.

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A Hyperparameters and Reproducibility

Both MARBERTv2 and CAMeLBERT-MSA were fine-tuned for 10 to 15 epochs using a learning rate of 2×10^{-5} alongside Early Stopping with a patience of 3. To accommodate the sliding window size of 512 tokens within GPU memory constraints, we utilized a per-device batch size of 8 paired with 2 gradient accumulation steps, resulting in an effective batch size of 16. Mixed precision (FP16) was employed to optimize training speed and memory utilization.

B Detailed Corpus Statistics

Analysis of the AraSeg Corpus revealed significant document length challenges, necessitating the sliding-window tokenization outlined in Section 3.2. Subword length analysis (using the CAMeLBERT tokenizer) across the provided splits yielded the following distributions:

- **Training Set (156 docs):** Mean length of 866.4 subwords. 99 documents (63.5%) exceeded the 512-token limit. The maximum document length was 22,692 subwords.
- **Development Set (18 docs):** Mean length of 1,213.4 subwords. 14 documents (77.8%) exceeded the limit. The maximum document length was 3,011 subwords.
- **Blind Test set (100 docs):** Mean length of 1,003.0 subwords. 63 documents (63.0%) exceeded the limit. The maximum document length was 19,406 subwords.